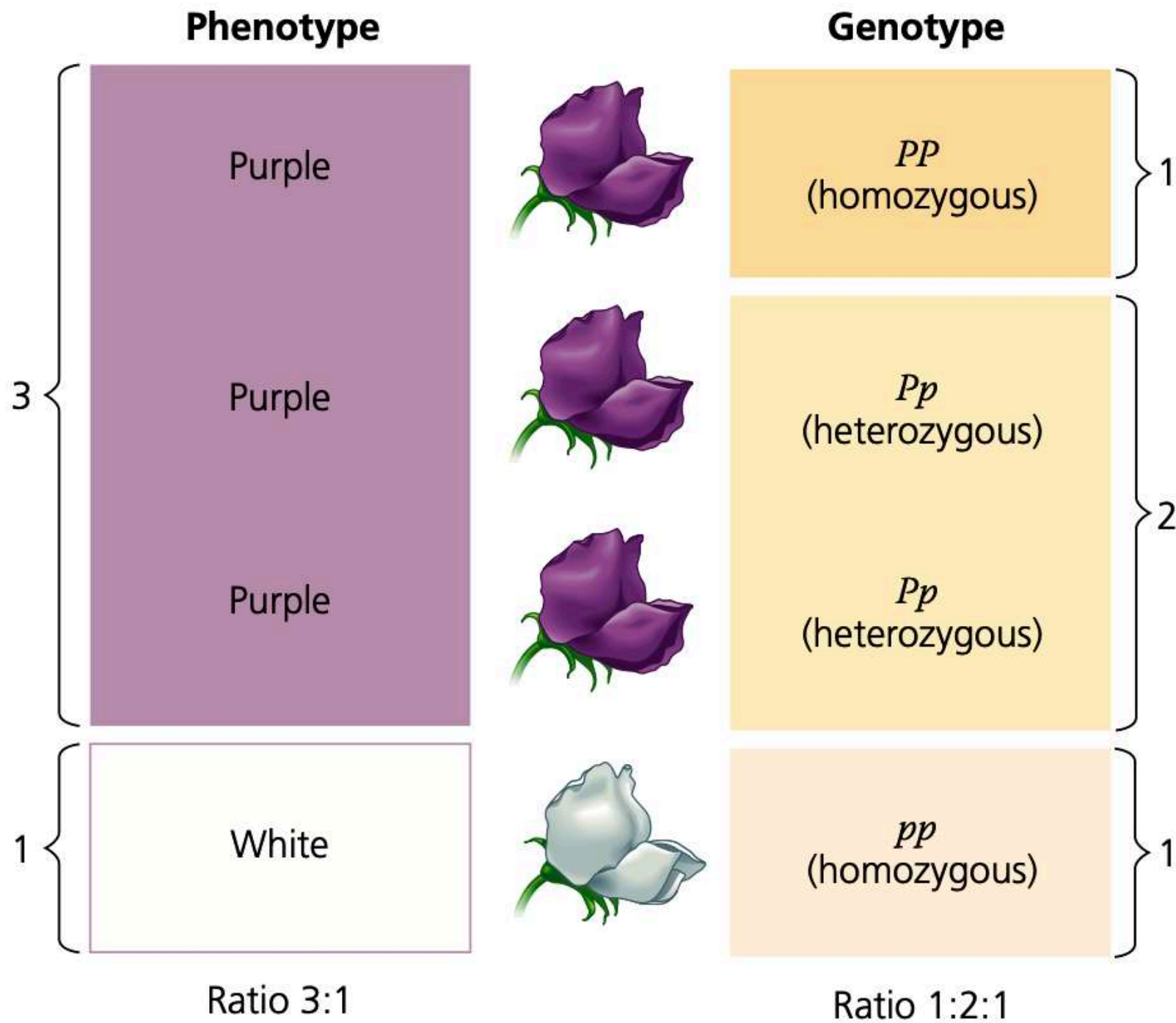


Law of Segregation



- **Phenotype** = organism's appearance or observable traits
- **Genotype** = genetic makeup or allele combinations
- Phenotype and Genotype ratios are **mathematically predictable**
- Mendel inferred these without knowing concepts of chromosomes or genes or meiosis or mitosis
- Homozygous **inbreeding** = offsprings will have genotype and phenotype of parent
- Heterozygous breeding = offsprings will have a mix of parents and grandparents genotype and phenotype

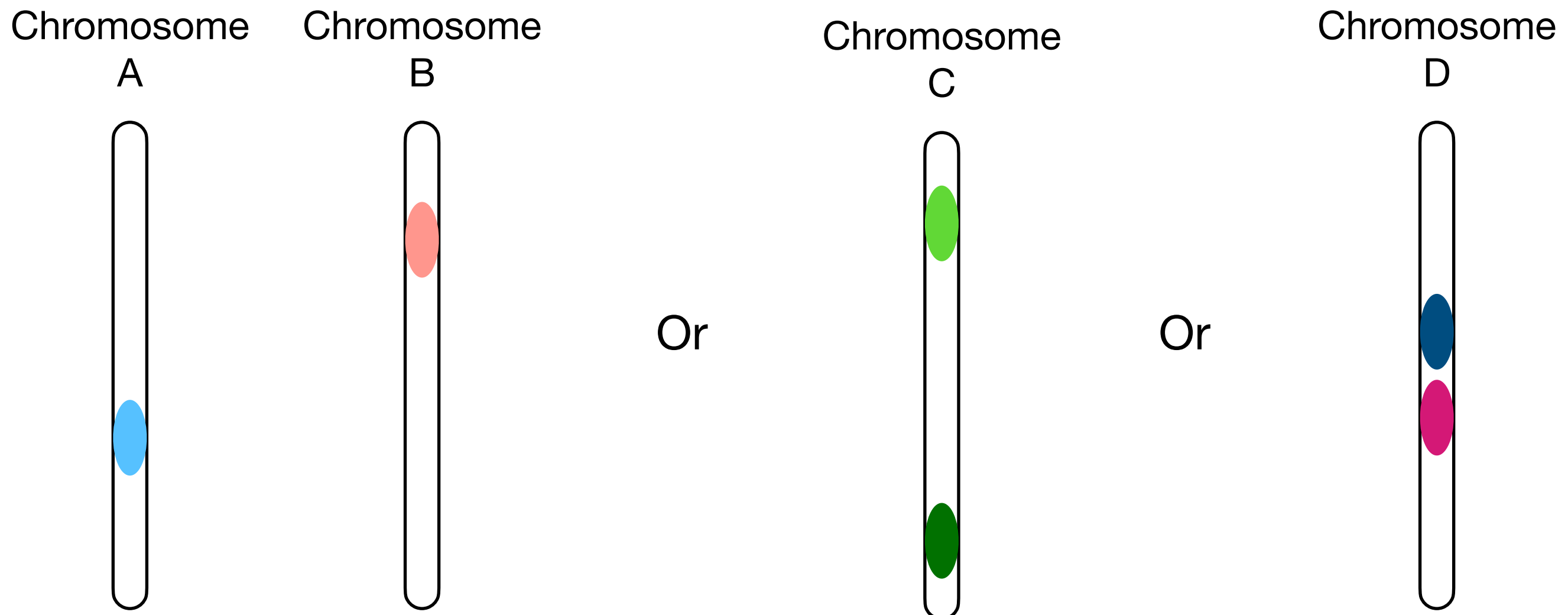
If you only know phenotype (e.g. purple flowers), it is impossible to be sure of the genotype (can be PP or Pp). Need to do a **Test cross** with a known true breeding PP and/or pp genotype and track offspring for multiple generations

Law of Independent Assortment

Mendel deduced the Law of Segregation by following the inheritance patterns of single traits controlled by one gene

What about 2 traits controlled by 2 genes or more?

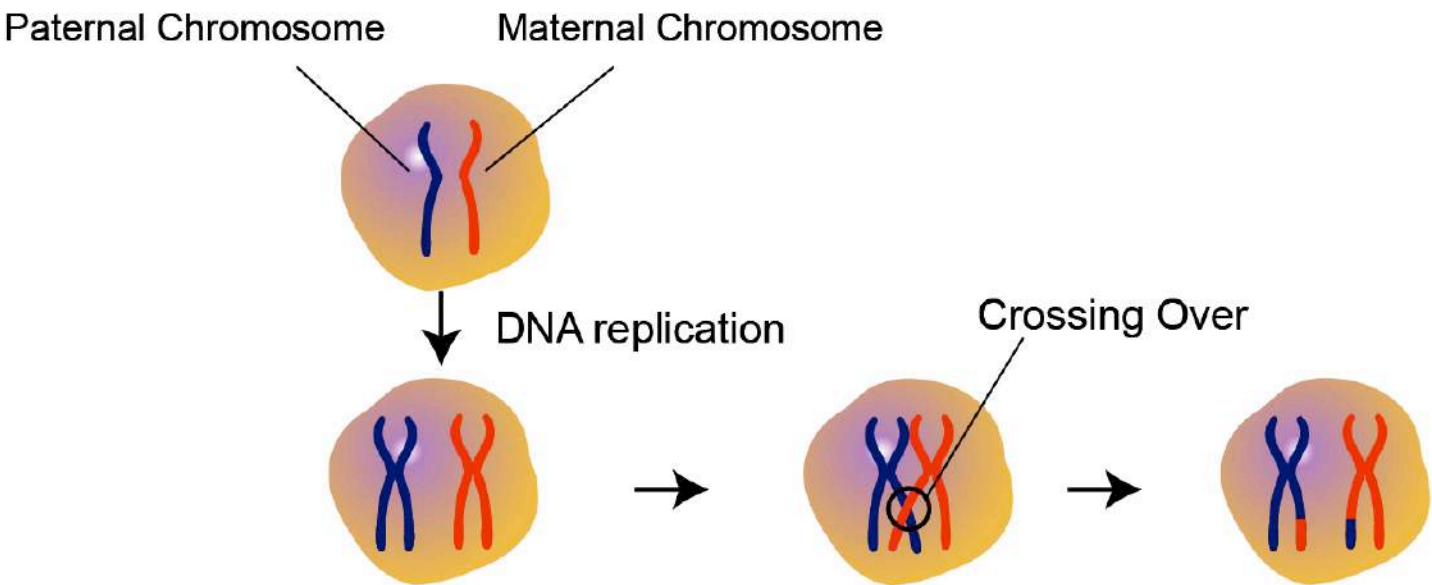
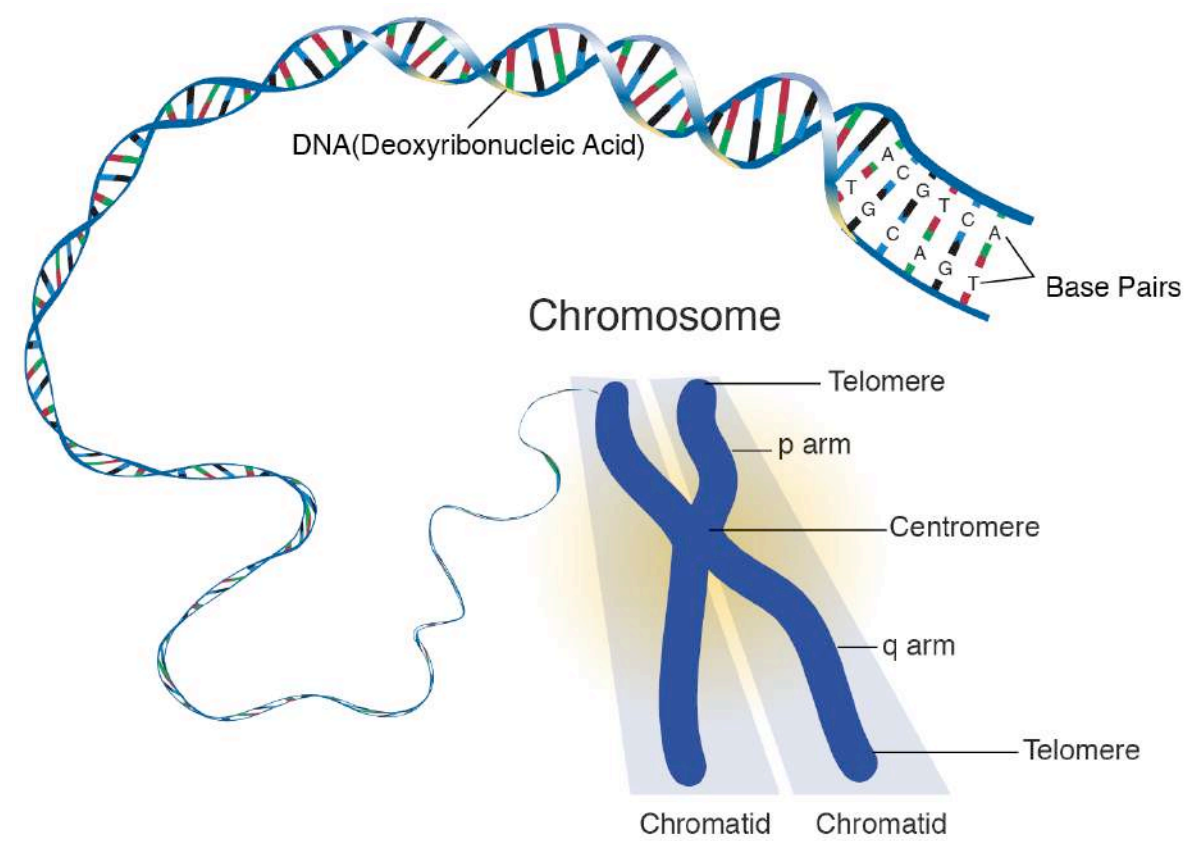
The 2 traits can be on different chromosomes or on the same chromosome and if on the same chromosome, they can be close or far from each other



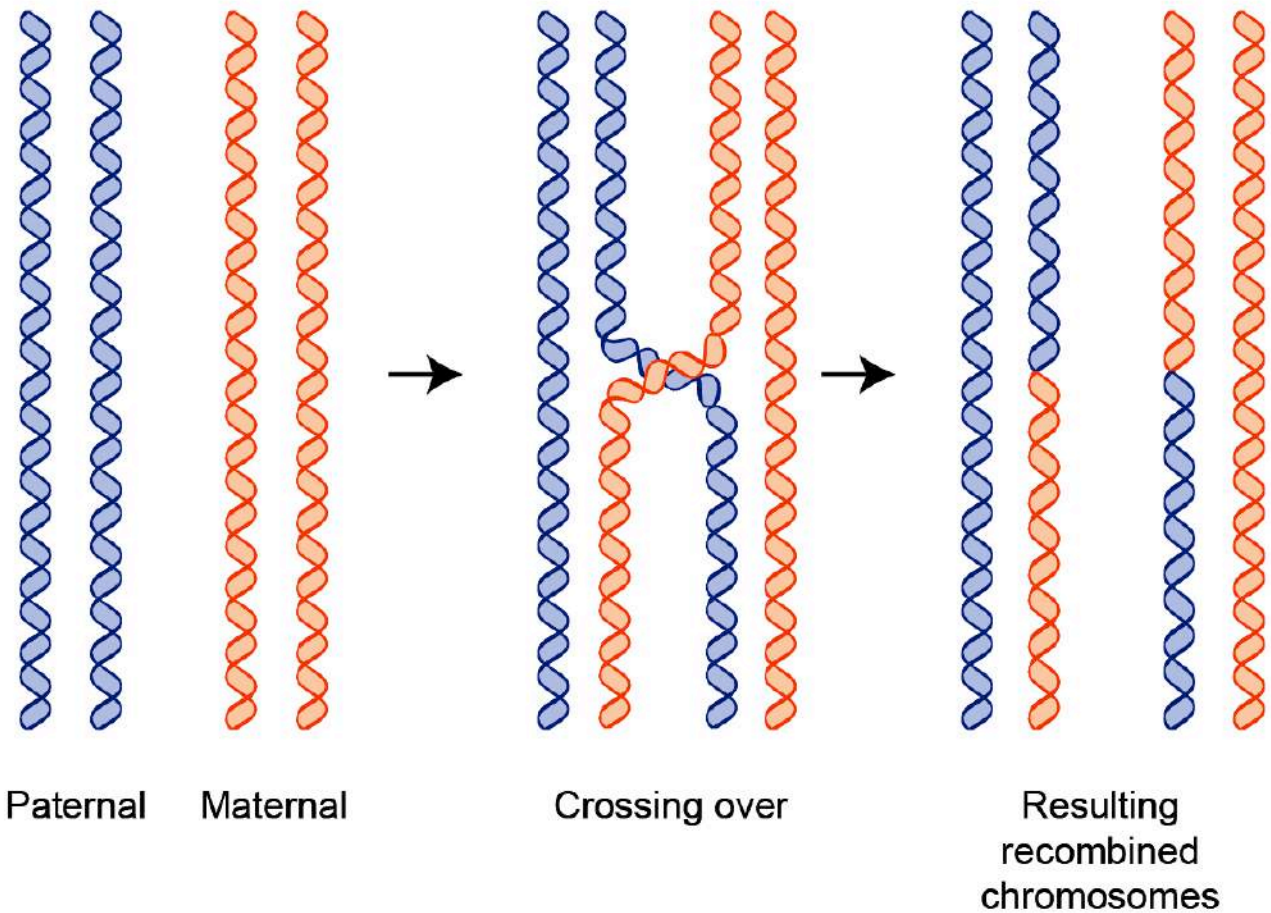
Only one chromatid shown in diagrams above. Ovals represent genes

Meiosis

Occurs in all species when sperm and eggs are produced
Basis of genetic diversity amongst individuals of a species



Recombination between 2 homologous chromosomes



Law of Independent Assortment

In nature multiple traits occur together: For example: short plant with round seeds, wrinkled and yellow

Mendel followed 2 traits at a time
Because of his monohybrid crosses, Mendel knew which traits were dominant and recessive

Example: Yellow or Green seed color and Round or Wrinkled seed shape

Table 14.1 The Results of Mendel's F ₁ Crosses for Seven Characters in Pea Plants					
Character	Dominant Trait	×	Recessive Trait	F ₂ Generation Dominant: Recessive	Ratio
Flower color	Purple 	×	White 	705:224	3.15:1
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1
Pod shape	Inflated 	×	Constricted 	882:299	2.95:1
Pod color	Green 	×	Yellow 	428:152	2.82:1
Flower position	Axial 	×	Terminal 	651:207	3.14:1
Stem length	Tall 	×	Dwarf 	787:277	2.84:1

Table 14.1 The Results of Mendel's F ₁ Crosses for Seven Characters in Pea Plants					
Character	Dominant Trait	×	Recessive Trait	F ₂ Generation Dominant: Recessive	Ratio
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1

Law of Independent Assortment

Yellow Seed = dominant = *Y*

Green seed = recessive = *y*

Round Seed = dominant = *R*

Wrinkled seed = recessive = *r*

YYRR

YYRR

Table 14.1 The Results of Mendel's F ₁ Crosses for Seven Characters in Pea Plants					
Character	Dominant Trait	×	Recessive Trait	F ₂ Generation Dominant: Recessive	Ratio
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1

yyRR

YYrr

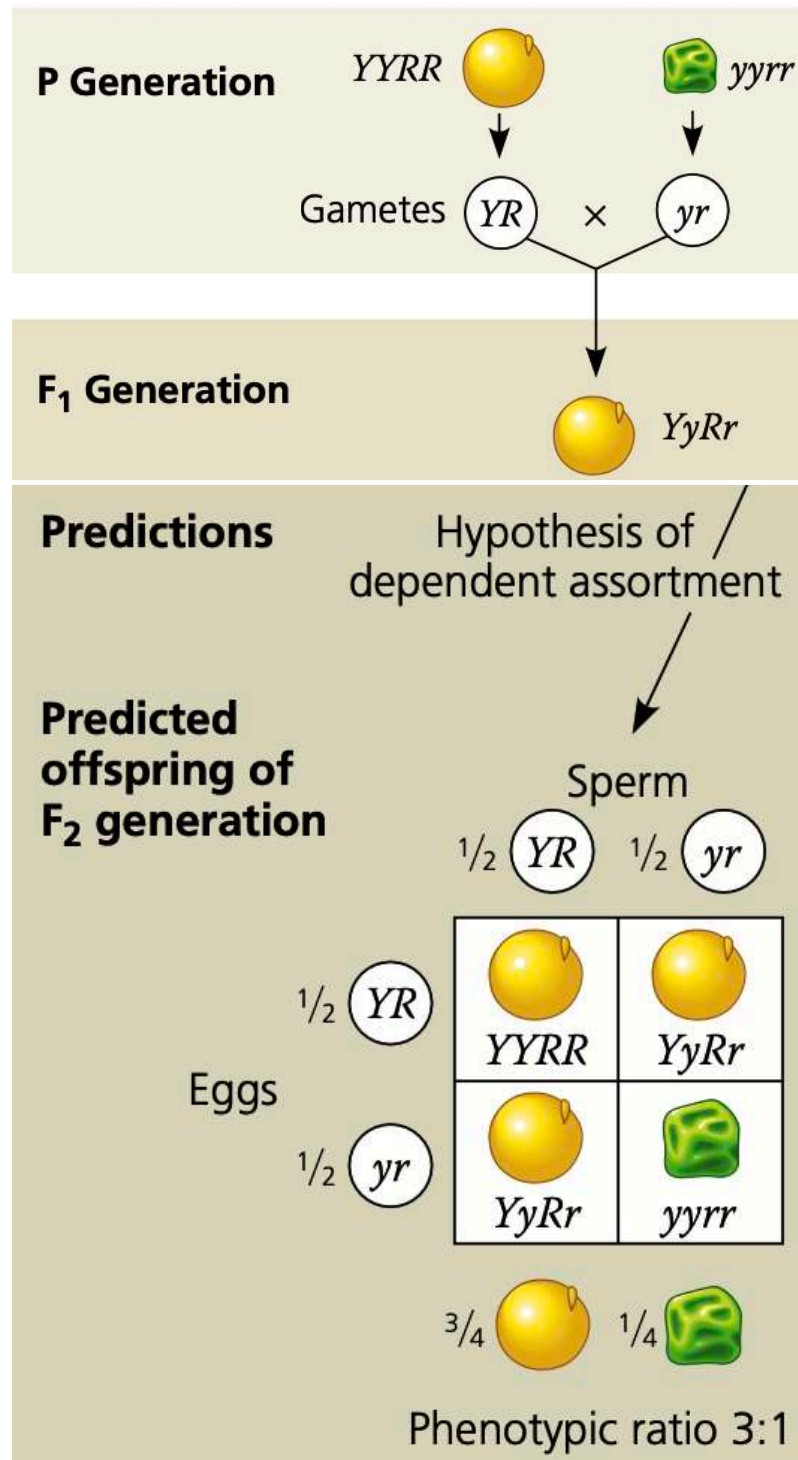
Write the diploid genotype for both traits for each of the seeds shown in the above picture

Law of Independent Assortment

Mendel crossed two true-breeding plants: yellow-round seeds x green-wrinkled seeds

Hypothesis being tested:

do true dominant or recessive traits always get inherited together?

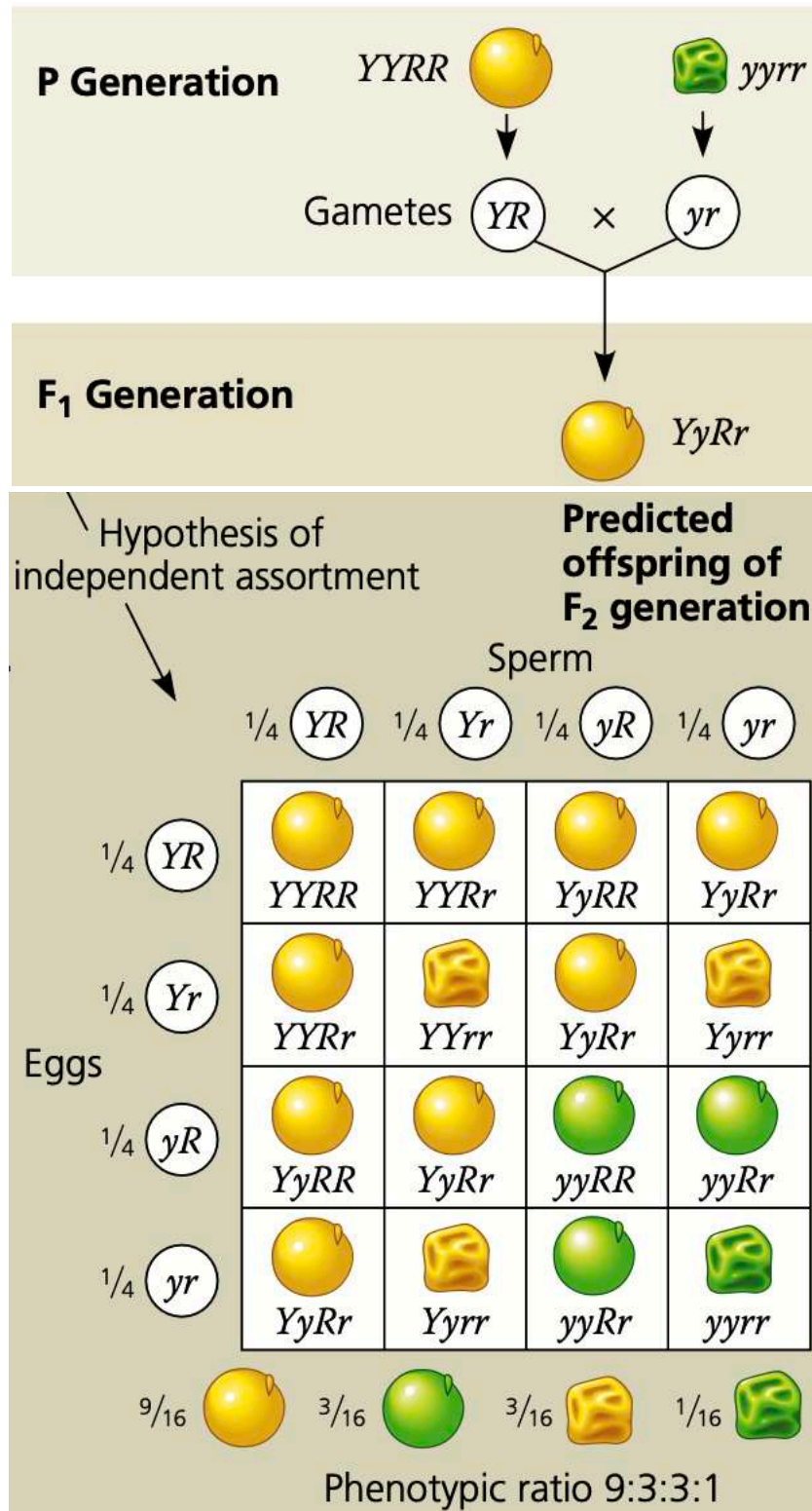


- Each **true breeding P plant** has one allele of the *Y* gene and one allele of *R* gene or one allele of the *y* gene and one allele of *r* gene. Each P plant has two copies of the allele (*YY* and *RR* or *yy* and *rr*) as it is a diploid
- Each **P plant** produces **one type of haploid gamete**, either, yellow round allele gamete (*YR*) **or** green wrinkled allele gamete (*yr*)
- When gametes from two P plants fuse, they can only form **one diploid combination in F₁ upon fertilization: YyRr**
- The cross-pollinated F₁ plant has two alleles for each trait, *Y* and *y*, *R* and *r*, but *Y* and *R* are dominant and therefore the F₁ plant has yellow round seeds.
- If each pair of dominant or recessive traits are to be inherited together:** Each **F₁ plant will have to produce two types of gametes: YR and yr**
- When gametes of F₁ plants fuse (self or cross-fertilization), they can form **three types of diploid combinations in F₂: YYRR or YyRr or yyrr**
1:2:1 genotype and 3:1 phenotype ratio

THIS IS NOT THE RESULT MENDEL OBTAINED

Law of Independent Assortment

Mendel crossed two true-breeding plants: yellow-round seeds x green-wrinkled seeds



- Each **true breeding P plant** has one allele of the Y gene and one allele of R gene or one allele of the y gene and one allele of r gene. Each P plant has two copies of the allele (YY and RR or yy and rr) as it is a diploid
- Each **P plant** produces **one type of haploid gamete**, either, yellow round allele gamete (YR) **or** green wrinkled allele gamete (yr)
- When gametes from two P plants fuse, they can only form **one diploid combination in F₁ upon fertilization**: $YyRr$
- The cross-pollinated F₁ plant has two alleles for each trait, Y and y , R and r , but Y and R are dominant and therefore the F₁ plant has yellow round seeds.
- If each pair of dominant or recessive traits could be inherited separately: Each **F₁ plant will** produce **four types of gametes**: YR , Yr , yR and yr
- When gametes of F₁ plants fuse (self or cross-fertilization), they can form **several types of diploid combinations in F₂**

Two new traits appear in F₂:
green round and yellow wrinkled
THIS IS THE RESULT MENDEL OBTAINED

Law of Independent Assortment

Mendel crossed two true-breeding plants: yellow-round seeds x green-wrinkled seeds

Two new traits appear in F₂:
green round and yellow wrinkled

This is ONLY POSSIBLE

If *Y* and *R* separated from each other

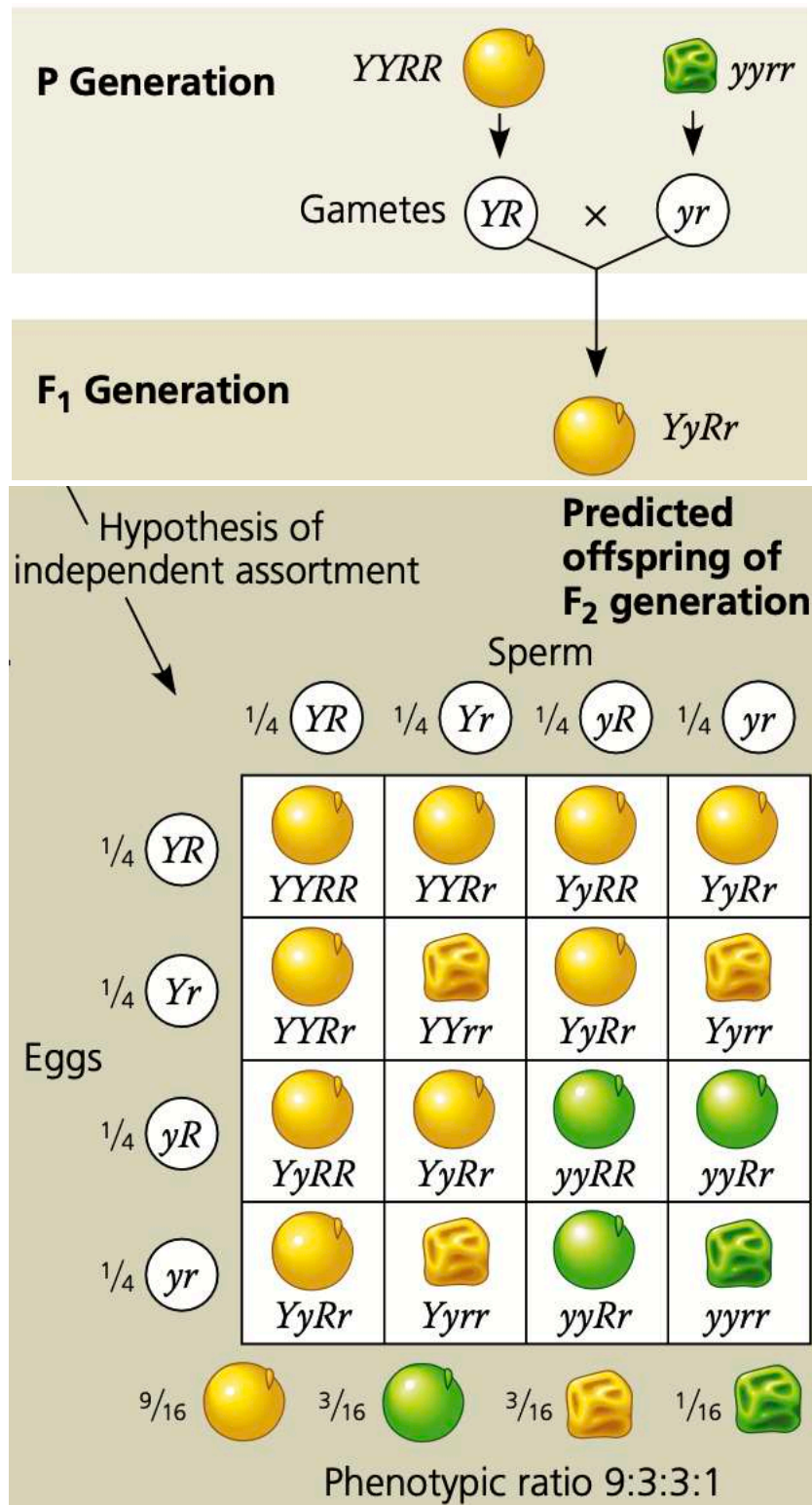
and

If *y* and *r* separated from each other

and

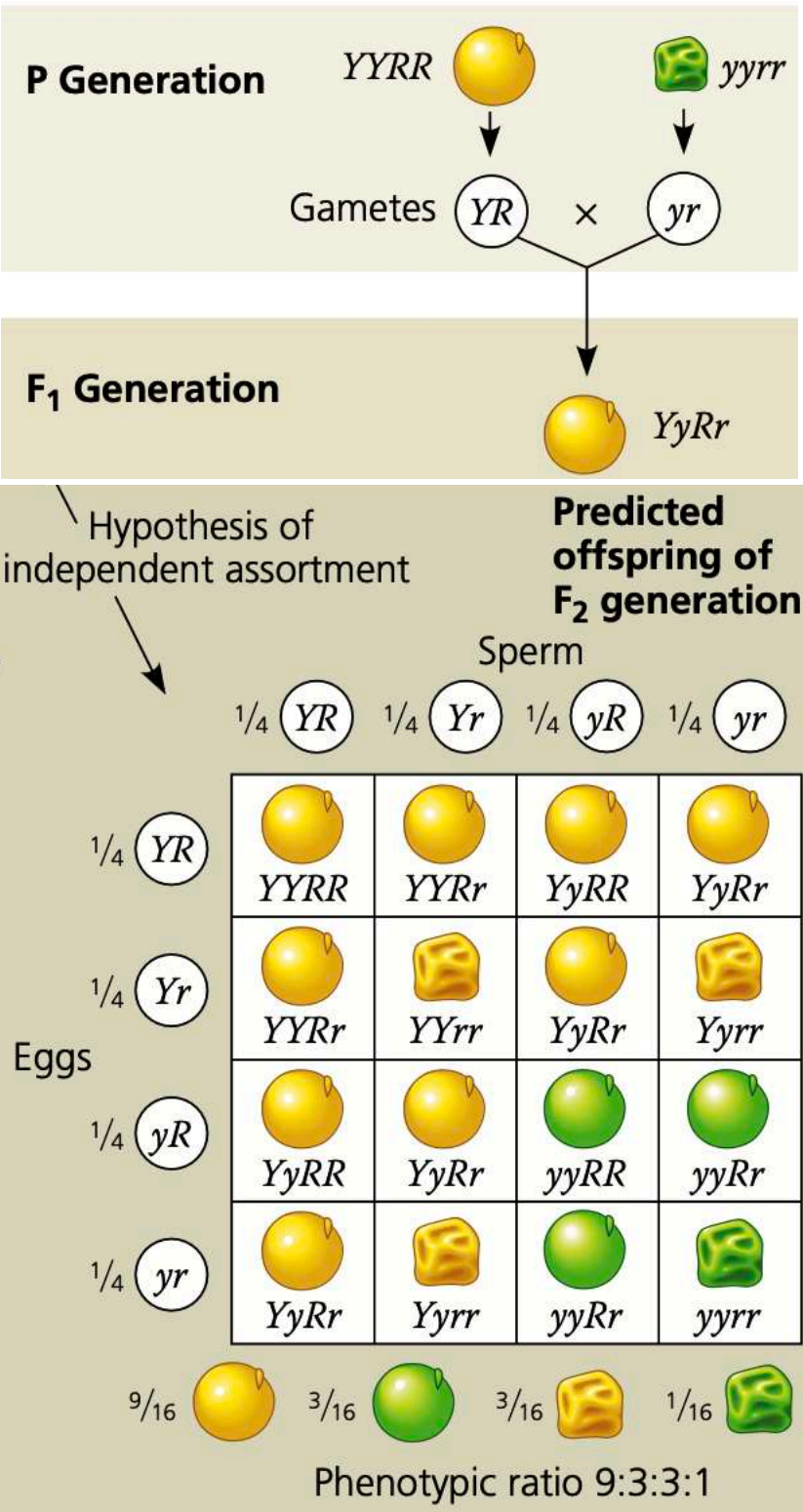
combined with each other

two or more genes assort independently -
that is, each pair of alleles segregates
independently of each other pair of alleles during
gamete formation



Law of Independent Assortment

Mendel crossed two true-breeding plants: yellow-round seeds x green-wrinkled seeds



two or more genes assort independently -
that is, each pair of alleles segregates independently of each other pair of alleles during gamete formation

applies only to genes (allele pairs) located on non-homologous chromosomes or, alternatively, to genes that are very far apart on the same chromosome

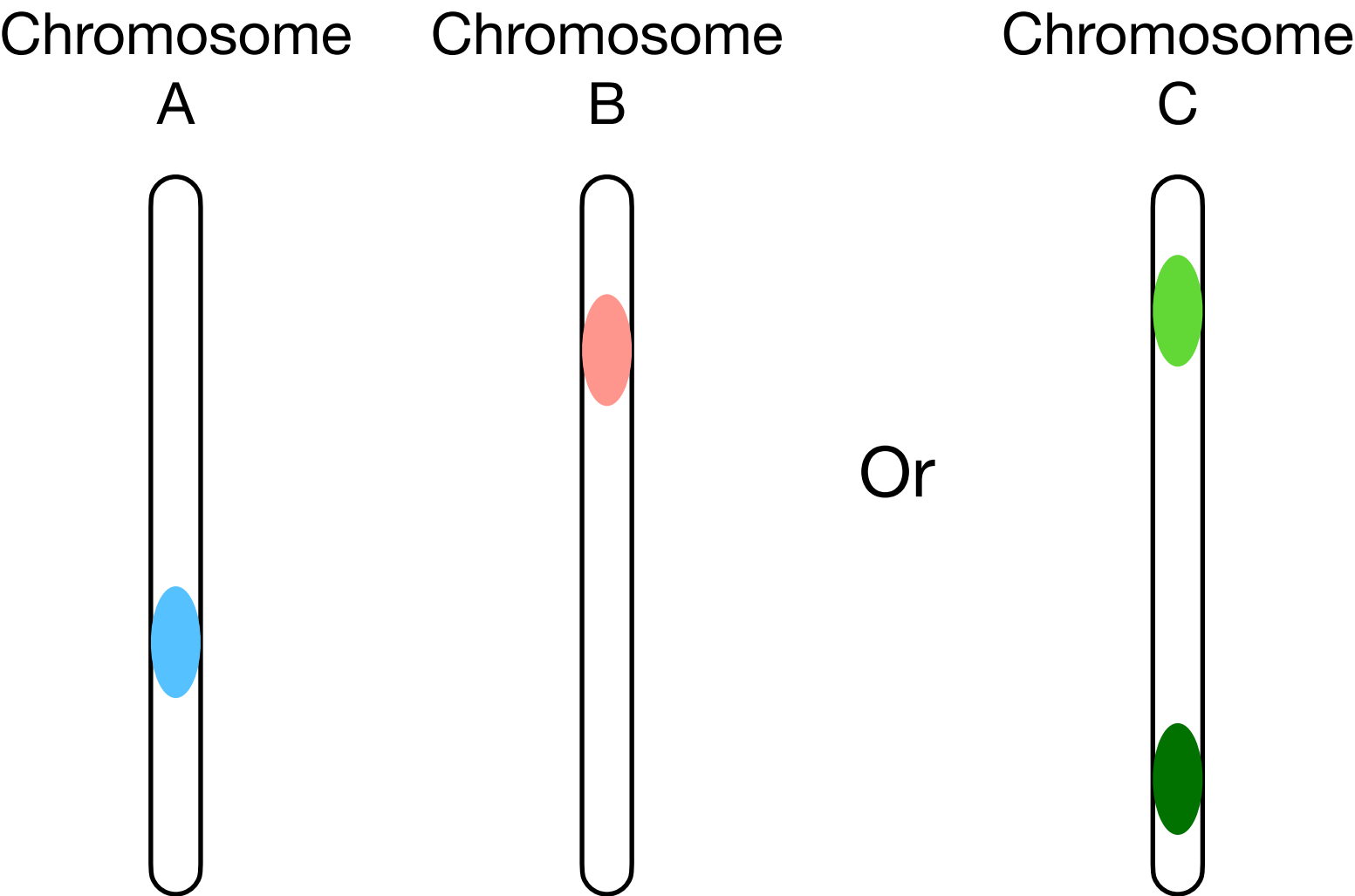
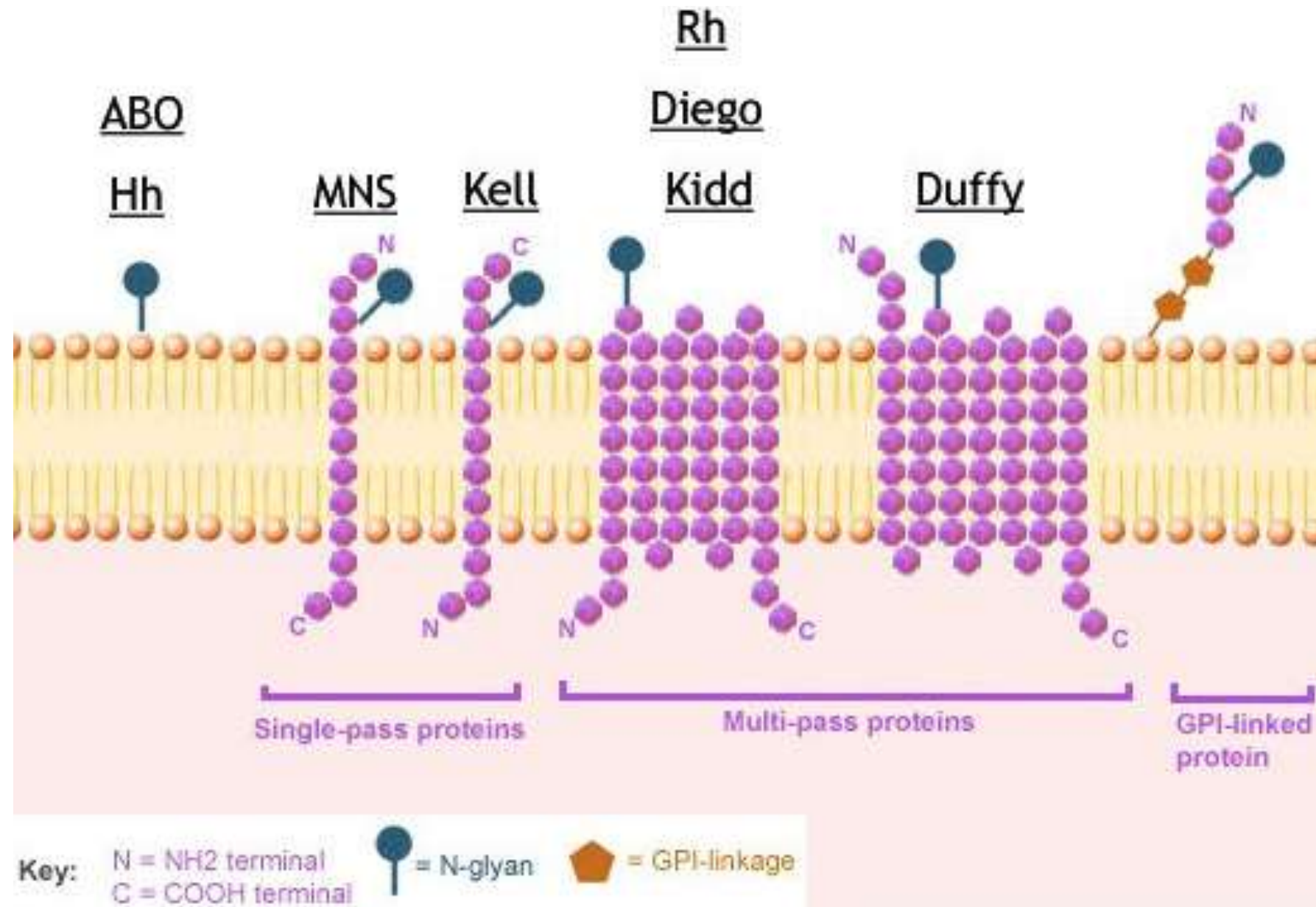
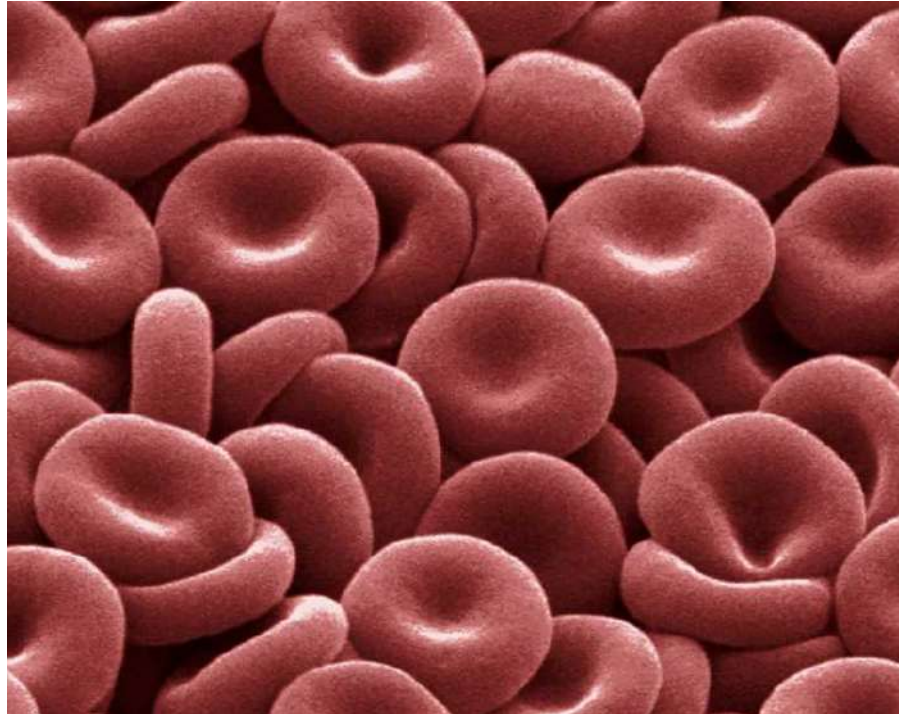


Figure 14.8 and 15.2 of Campbell's Biology: a global approach

Law of Independent Assortment

Surface of RBCs have many antigens: carbohydrates, glycoproteins, proteins etc



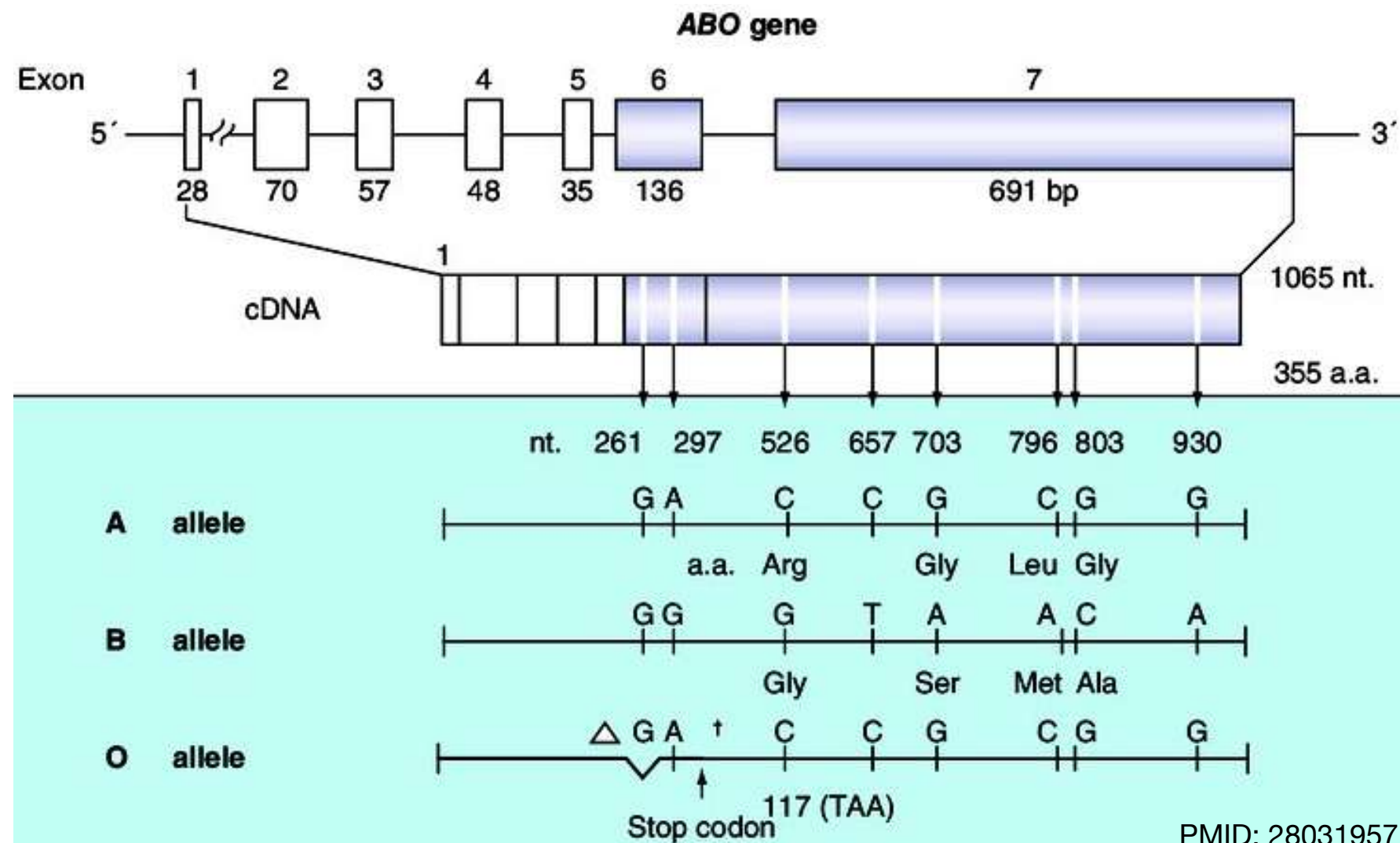
- One such antigen is the “A” or “B” antigen, which is a carbohydrate
- The sequence of the oligosaccharide determines if the antigen is A or B
- The oligosaccharide chain is attached to proteins and lipids in the RBC membrane
- Second such antigen is the “Rh” antigen, which is protein (49 Rh variations exist)

Law of Independent Assortment

The ABO gene indirectly encodes the ABO blood group antigens

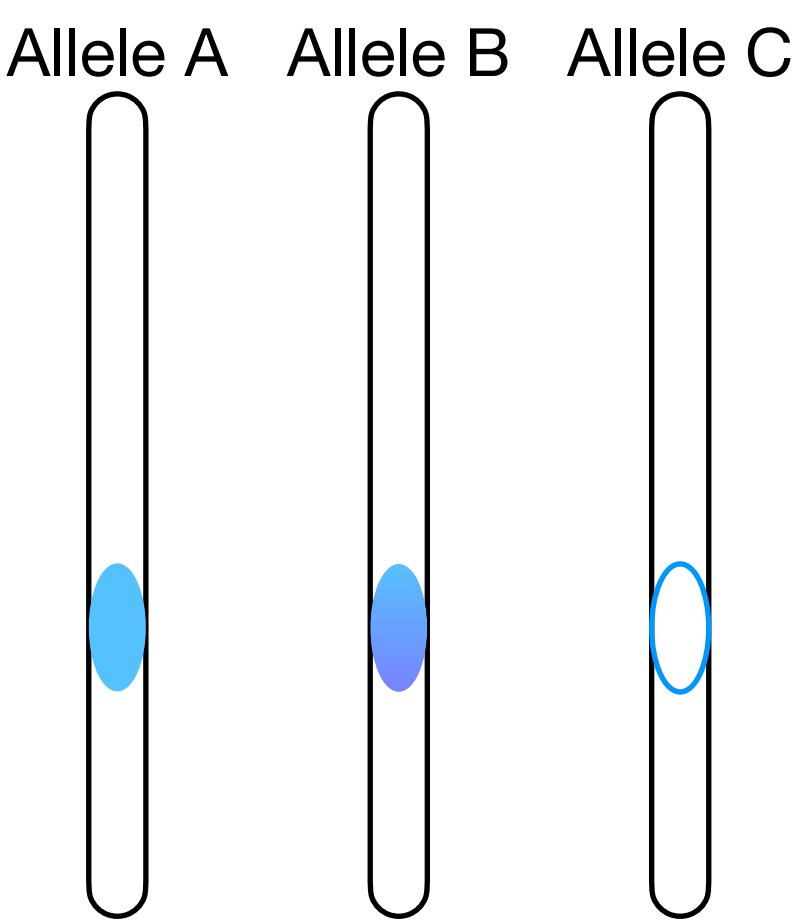
- The ABO locus has three main allelic forms: A, B, and O
- The A and B alleles each encode a glycosyltransferase that converts an antigen precursor, H, into the A and B antigen
- A or B allele arises from a few changes in the DNA sequence = A and B enzymes that differ by four amino acids
- O allele has a change in the DNA sequence which introduces a stop codon = inactive enzyme which cannot convert the precursor, H, into the A and B antigen

A allele = dominant
B allele = also dominant
O allele = recessive



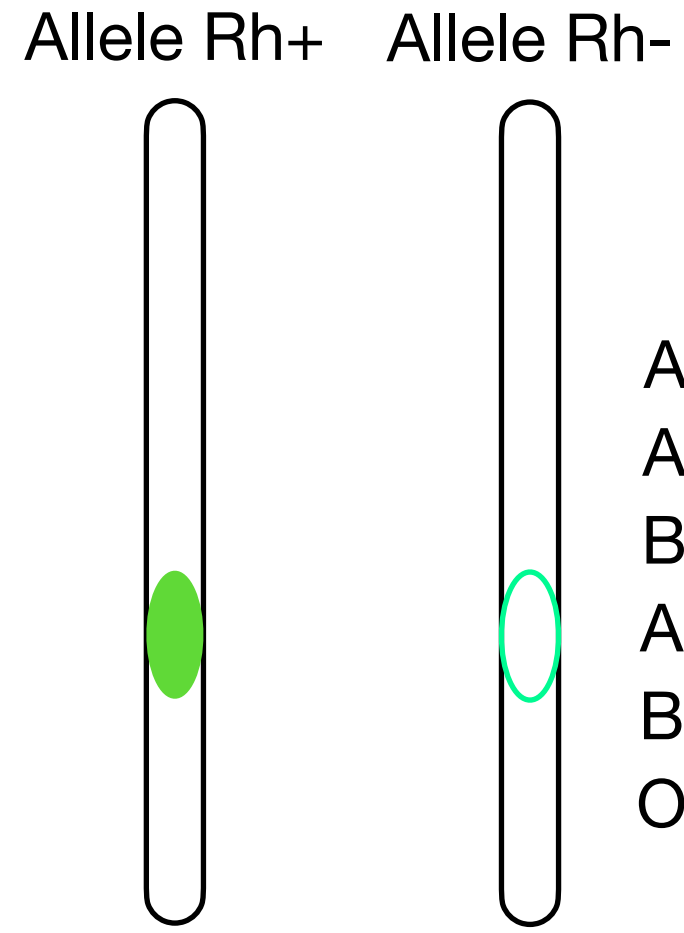
Law of Independent Assortment

Chromosome 9



A allele = dominant
B allele = also dominant
O allele = recessive

Chromosome 1



Rh+ allele = dominant
Rh- allele = recessive

Possible diploid genotype combinations?

AARh+Rh+	AARh-Rh-	AARh+Rh-
ABRh+Rh+	ABRh-Rh-	ABRh+Rh-
BBRh+Rh+	BBRh-Rh-	BBRh+Rh-
AORh+Rh+	AORh-Rh-	AORh+Rh-
BORh+Rh+	BORh-Rh-	BORh+Rh-
OORh+Rh+	OORh-Rh-	OORh+Rh-

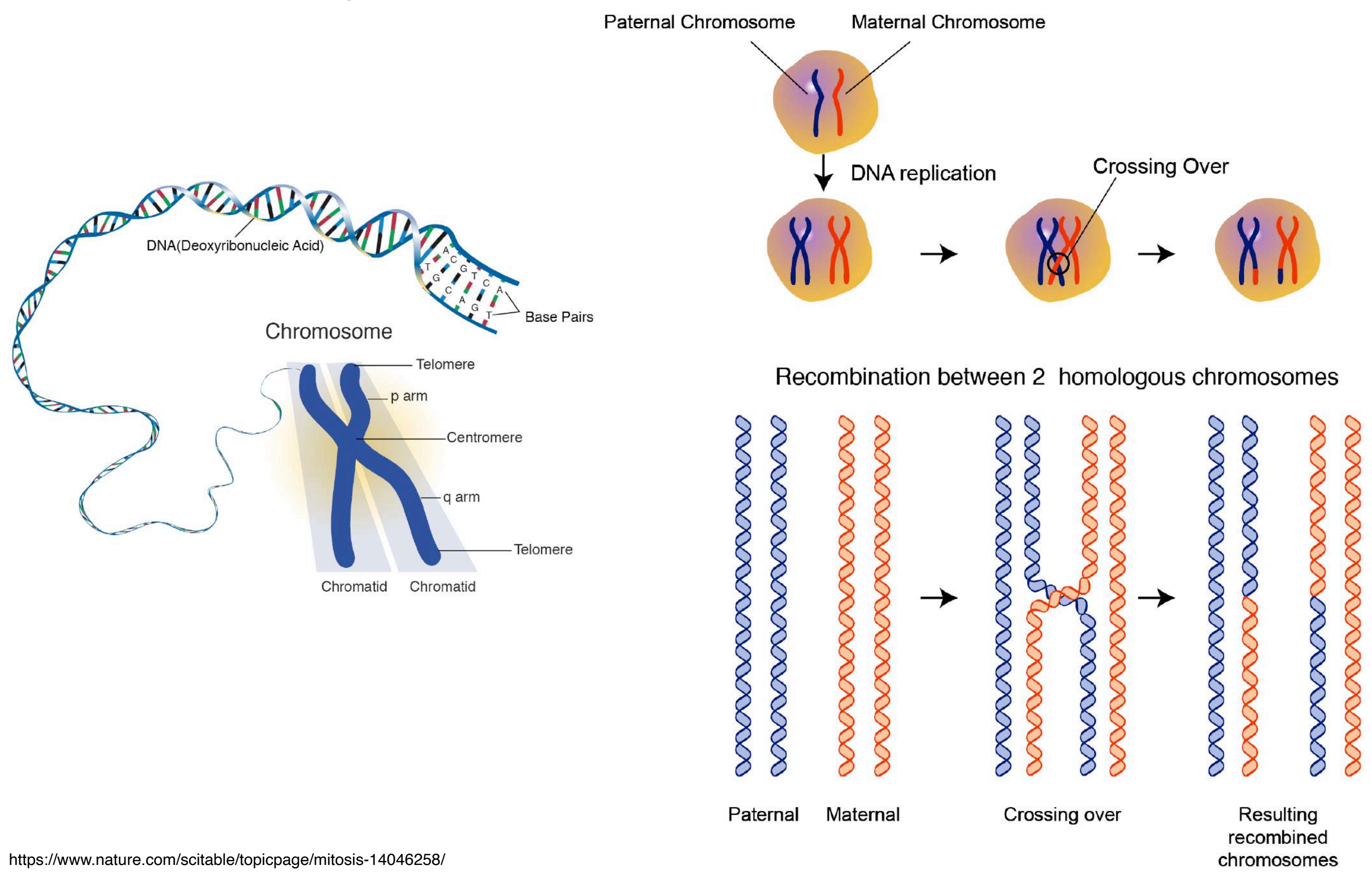
AB/BA, AO/OA and BO/OB
considered as one each above

Which of the genotypes above would be similar to the P generation yellow round x green wrinkled cross that Mendel did?

Only one chromatid shown in diagrams above to show what is possible in a gamete

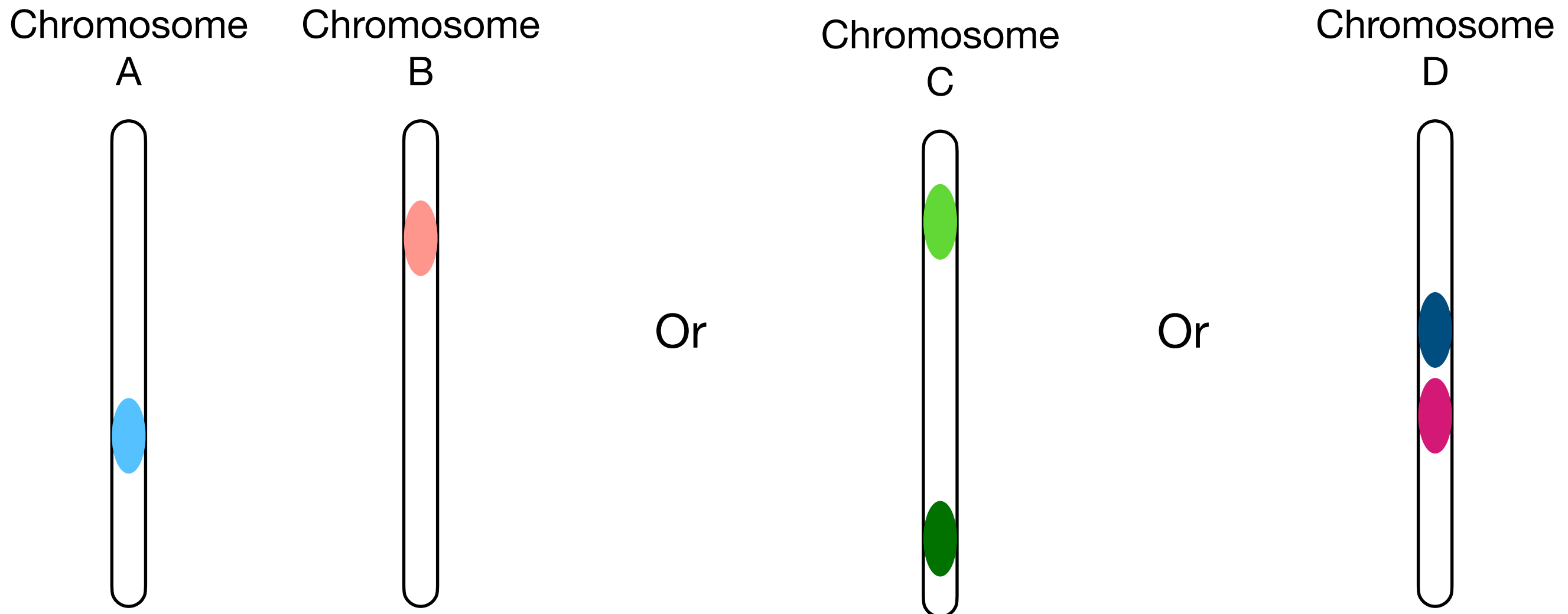
Relationship among alleles of a single gene	Description	Example
Complete dominance of one allele	Heterozygous phenotype same as that of homozygous dominant	PP  Pp 
Incomplete dominance of either allele	Heterozygous phenotype intermediate between the two homozygous phenotypes	 $C^R C^R$  $C^R C^W$  $C^W C^W$
Codominance	Both phenotypes expressed in heterozygotes	$I^A I^B$ 
Multiple alleles	In the population, some genes have more than two alleles	ABO blood group alleles I^A, I^B, i
Pleiotropy	One gene affects multiple phenotypic characters	Sickle-cell disease

Fundamental importance of recombination in the context of gene positions on a chromosome



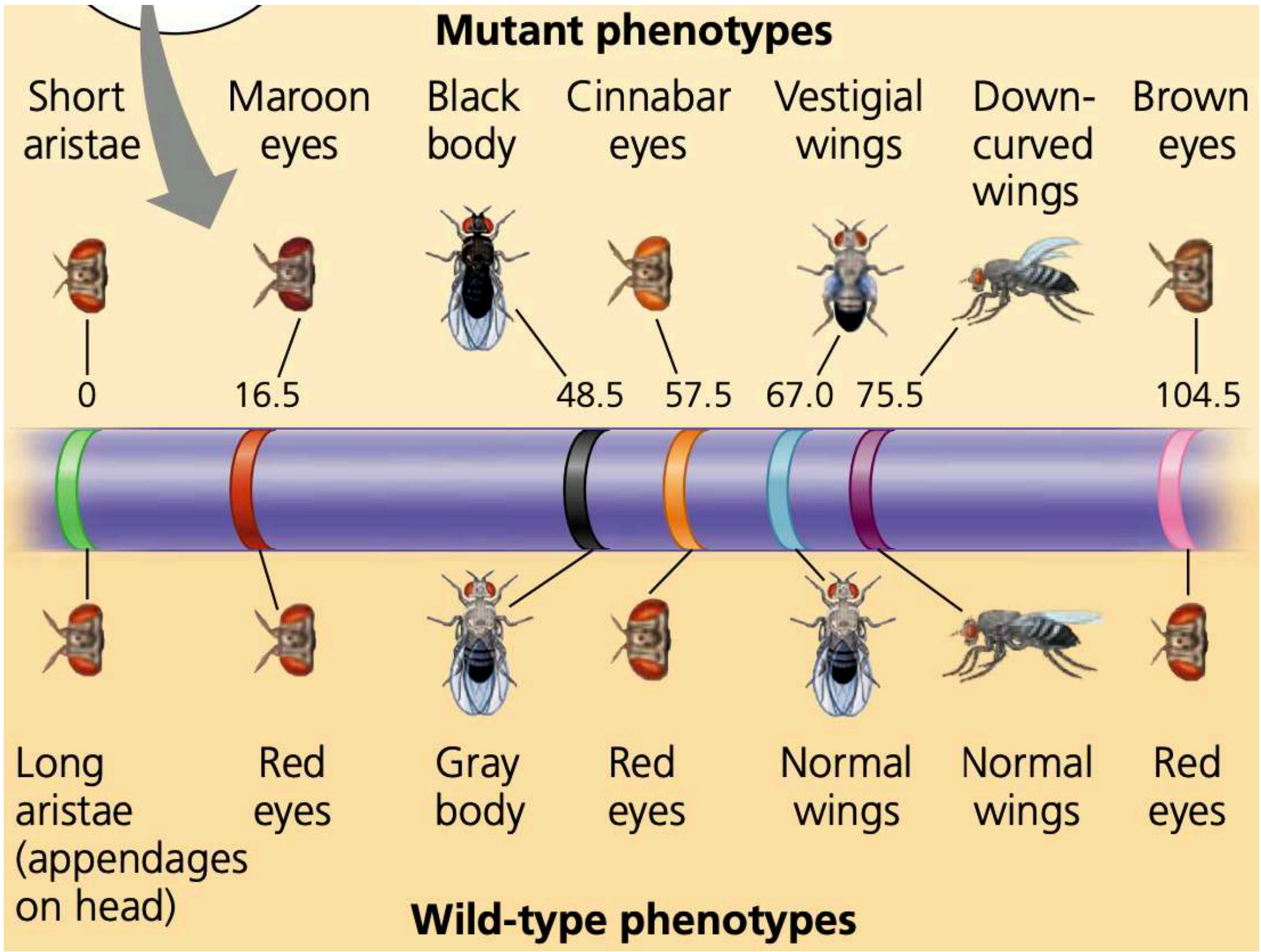
Genetic Linkage

- Genes on different chromatids are not physically contiguous on the same DNA molecule and can be inherited independently
- Genes on same chromatid are physically contiguous on the same DNA molecule and can be inherited independently, **if they are far from each other**
- Genes on same chromatid are physically contiguous on the same DNA molecule and **are likely be inherited independently extremely rarely, if they are near to each other.**
- **Closer they are to each other, the less likely it is for them to be inherited independently**



Only one chromatid shown in diagrams above. Ovals represent genes

Genetic Linkage



If a cross between two individuals produced 100 progeny, in how many of these progeny do two traits manifest together?

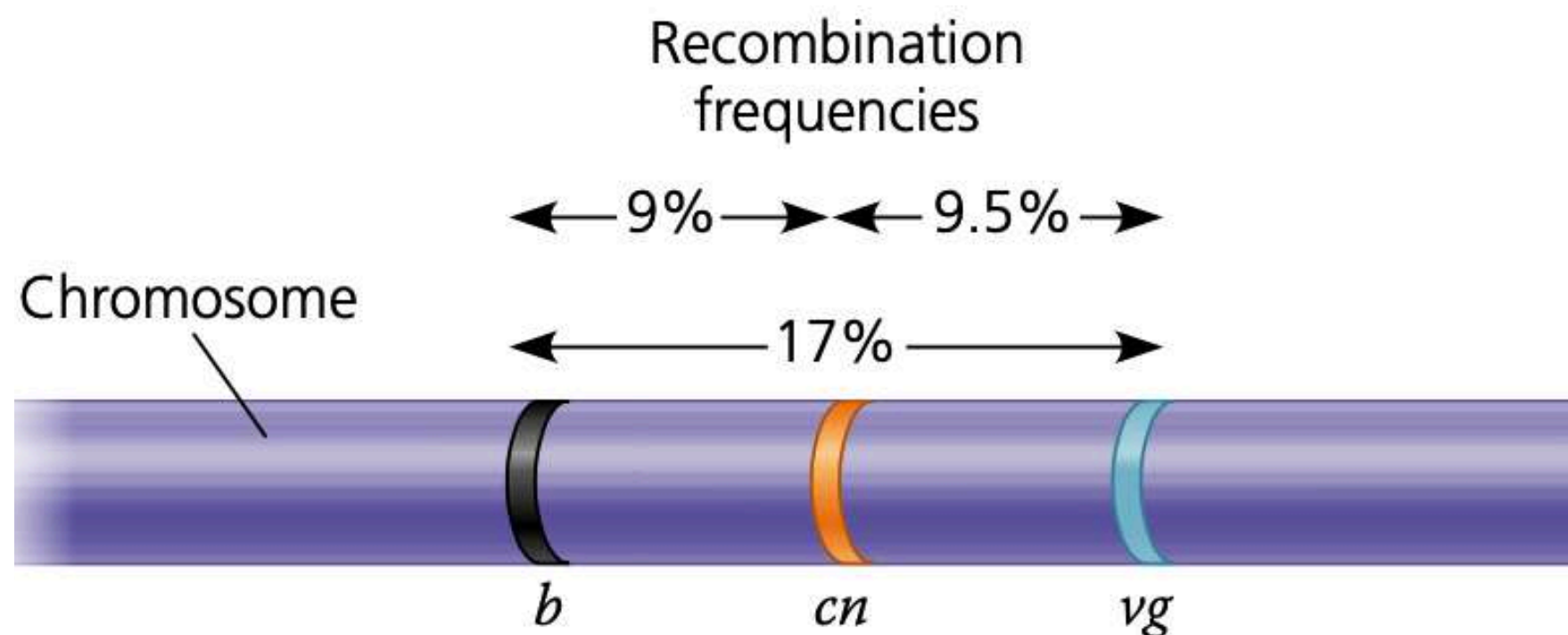
Depends on how close 2 genes are on the same chromosome

The closer the genes are, more likely they will be inherited together or not recombined away

Genetic Linkage

How does one estimate the distances between genes on chromosomes?

Typically if traits are separated (recombined) in 1 out of 100 progeny (1%), then the genes controlling those traits are 1 centimorgan (cM) apart



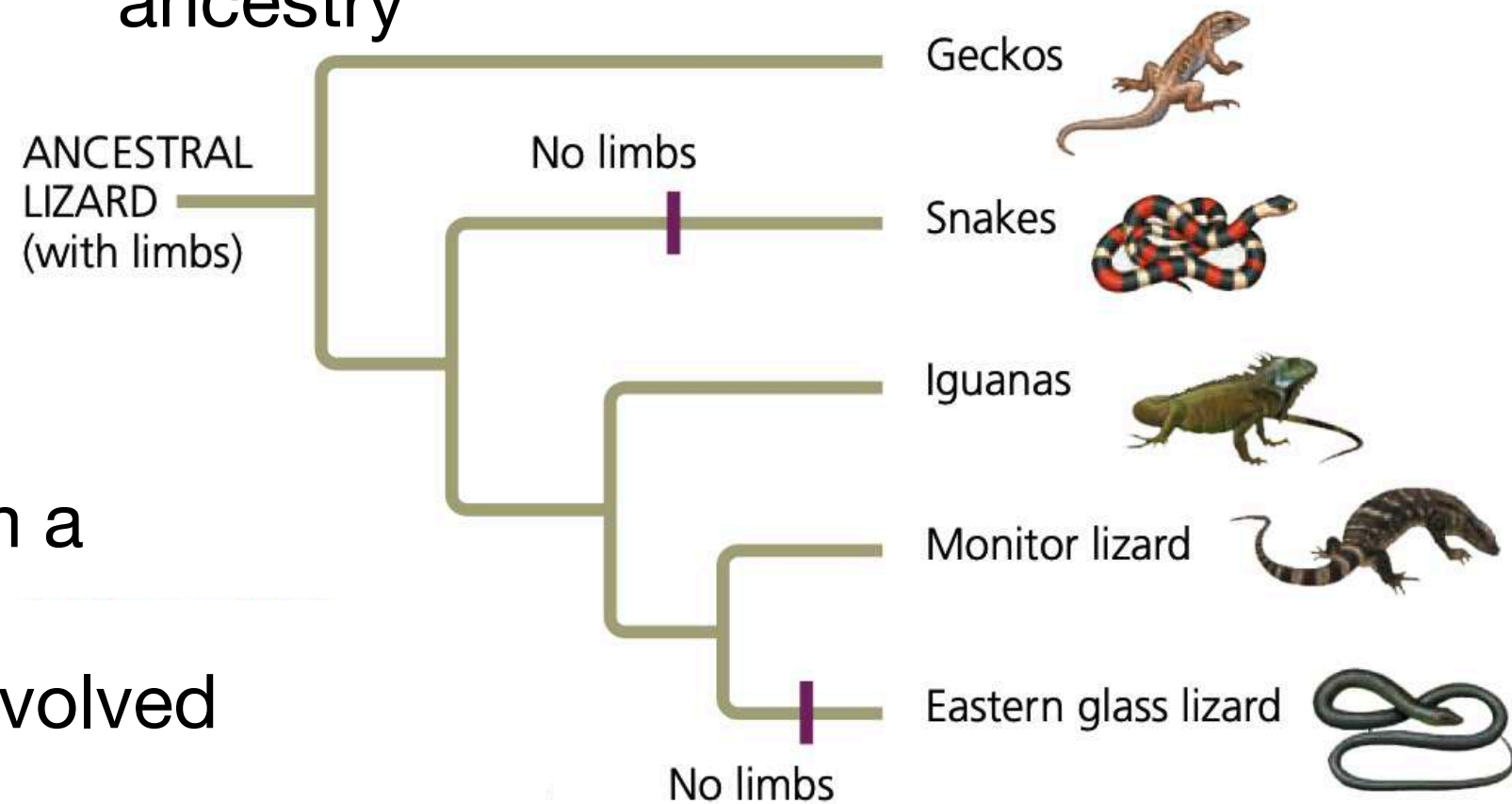
Ignore X-inactivation sections in Chapter 15

Evolution

Convergent vs Divergent evolution



Convergent: same traits evolve independently multiple times - different ancestry

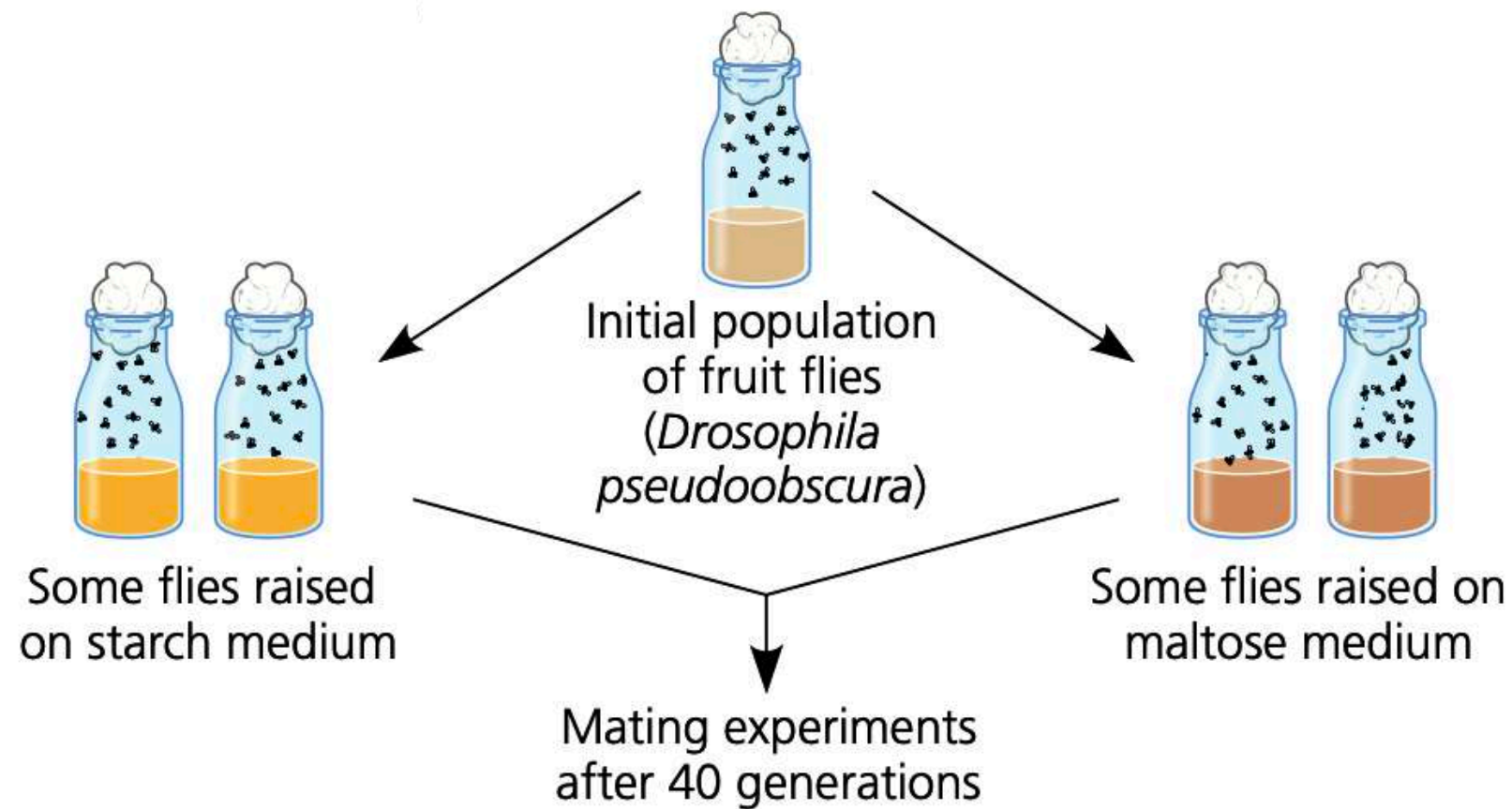


Divergent examples:

- All dog species evolved from a common wolf ancestor
- Elephants and Mammoths evolved from a common ancestor

Evolution

Reproductive barrier - defines a species



- Flies grown only on maltose or starch prefer to mate with like-partners
- Over the course of time, the two types of flies can evolve into two different species
- Groups experience different selection pressures - this may change physiology, behavior, genes etc leading to reproductive isolation
- Groups may also be separated geographically, leading to reproductive isolation

		Female	
		Starch	Maltose
Male	Starch	22	9
	Maltose	8	20

Number of matings in experimental group

		Female	
		Starch population 1	Starch population 2
Male	Starch population 1	18	15
	Starch population 2	12	15

Number of matings in control group